

Assessment - 2 Group Presentation

S226 HIT140 FOUNDATIONS OF DATA SCIENCE

Presented by:

Rupesh Ghimire (S403354)

Divya Malla (S406481)

Gaurav Aryal (S403279)

Quoc An Vuong (S406838)

Submitted on: Sep 10, 2026

Task 1:

Do forwards and midfielders have significantly different average assists per 90 minutes?

Presentator: Divya Malla (S406481)

Data Wrangling

1

Clean

Removed fully empty rows
Checked missing values



2

Select

Player
Position
Assists per 90



3

Transform

Renamed Ast.1
to Assists_per_90



4

Filter

Kept exact FW
and MF positions only

After wrangling: 82 forwards | 162 midfielders | 0 missing assists/90 values

Sampling and Statistical Test

Sampling

Random sample of 40 forwards.

Random sample of 40 midfielders.

Total sample = 80 players.

Same random sample can be reproduced.

Test used

Welch's two-sample t-test.

Used to compare the means of two independent groups.

95% confidence interval for FW – MF.

H_0 : the two population means are equal.

Result & Interpretation

FW mean

0.032

MF mean

0.0678

p-value

0.327

95% confidence interval

FW – MF: -0.1081 to 0.0366

The interval includes 0.

Conclusion

Since the p-value (0.327) is greater than 0.05, we do not reject H_0 . There is not enough evidence to conclude that forwards and midfielders have significantly different average assists per 90 minutes

Task 2:

Do winning teams hold significantly greater ball possession than losing teams?

Presenter: Quoc An Vuong (S406838)

Task 3:

Shots on Target: Group stage vs Knockout stage.

Presentator: Gaurav Aryal (S403279)

Task 4:

Goal Conversion Efficiency: Home vs Away Match

Presentator: Rupesh Ghimire (s403354)

Question formation and Data Collection

Since in any game, there is mostly home advantage to home teams. The home crowd, home jersey, familiar pitch and all other factors play a key role to team's performance. For example, Liverpool's home ground Anfield is considered one of the toughest ground to play against Liverpool.

Overall away game is tougher but:

IS THERE ANY SIGNIFICANT DIFFERENCE IN SHOT CONVERSION BETWEEN HOME AND AWAY GAMES?

To see this in FIFA World Cup 2026 matches, the data was obtained from the provided link of: <https://fbref.com/en/>

NOTE: An excel file was imported which was saved to csv and used in notebook.

PS: In the world cup, there isn't home and away match as per stadium but this is done only for analysis as each match has home and away teams and respective jersey which can also play importance in team's performance.

Data Wrangling

What was done:

- Columns selection
- Missing Value check and filter
- Data filtering for the use case / Feature Engineering
- Population and sample

Population = all usable matches (n=95, after excluding zero-SOT rows).

Sample = a subset of population (simple random sample), n=40, fixed seed for reproducibility.

```
SEED = 42
SAMPLE_N = 40
sample = usable.sample(n=SAMPLE_N, random_state=SEED).reset_index(drop=True)
sample.head()
```

	home_team_name	away_team_name	home_team_goal_count	away_team_goal_count	home_team_shots_on_target	away_team_shots_on_target	home_conv	away_conv	diff
0	Brazil	Japan	2	1	7	2	0.285714	0.50	-0.214286
1	Uzbekistan	Colombia	1	3	2	4	0.500000	0.75	-0.250000
2	France	Sweden	3	0	12	3	0.250000	0.00	0.250000
3	Mexico	Ecuador	2	0	3	1	0.666667	0.00	0.666667
4	Mexico	South Africa	2	0	4	2	0.500000	0.00	0.500000

Descriptive Statistics

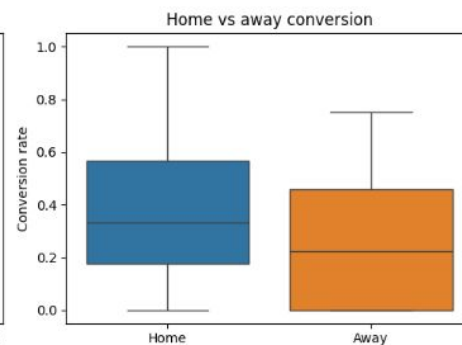
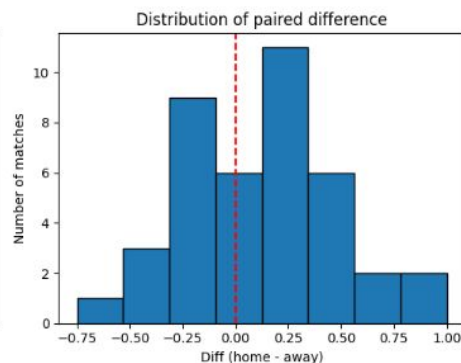
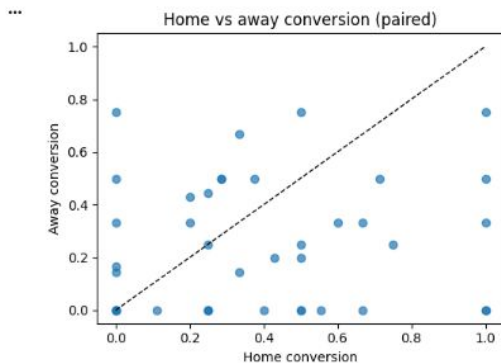
Describes the data through different metrics.

- Mean
- Median
- Standard Deviation
- Variance
- Quartiles
- Minimum
- Maximum
- Number of Samples

```
--- Home conversion ---  
n      : 40  
mean   : 0.3914  
median : 0.3333  
sd     : 0.3213  
variance : 0.1033  
Q1     : 0.1778  
Q3     : 0.5667  
min    : 0.0000  
max    : 1.0000
```

```
--- Away conversion ---  
n      : 40  
mean   : 0.2515  
median : 0.2250  
sd     : 0.2468  
variance : 0.0609  
Q1     : 0.0000  
Q3     : 0.4583  
min    : 0.0000  
max    : 0.7500
```

```
--- Diff (home - away) ---  
n      : 40  
mean   : 0.1399  
median : 0.2024  
sd     : 0.3812  
variance : 0.1453  
Q1     : -0.1488  
Q3     : 0.3500  
min    : -0.7500  
max    : 1.0000
```



Inferential Statistics

Inferential statistics uses data from a sample to make conclusions about a larger population.

Before doing any t-test, we need to do normality check because t-test assumes data follows normal distribution.

- Confidence Interval: A confidence interval gives a range of plausible values for a population parameter based on the sample.
- Hypothesis testing is a statistical method used to determine whether there is enough evidence to reject a null hypothesis.
- Although my research question compares home and away teams, the observations come from the same matches. Therefore, they are naturally paired. I calculated the home-minus-away difference for each match and used a one-sample t-test to determine whether the mean difference was significantly different from zero.
-

Conclusion (Task 4)

Result	Value 
Home mean	39.14%
Away mean	25.15%
Mean difference	13.99 percentage points
95% CI	1.80% – 26.18%
t-statistic	2.3218
p-value	0.0256